

Analytic State-Averaged DMRG-CASSCF Gradients and Nonadiabatic Couplings in PySCF with SHARC Integration

Author list to be finalized

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Abstract

We present an open-source PySCF/pyblock2 implementation of analytic state-averaged DMRG-CASSCF gradients and nonadiabatic couplings, together with a SHARC-compatible interface for nonadiabatic dynamics workflows. The implementation exposes both a native MPS response path and a PySCF-compatible FCI-solver wrapper, enabling validation against standard CASSCF gradient and derivative-coupling machinery on small and medium active spaces. We quantify the bond-dimension orbital-response error on public benchmark systems and demonstrate that DMRG-CASSCF energies, dipoles, gradients, and derivative couplings can be emitted in SHARC format. The present SHARC path uses FCI-projected response vectors and is therefore intended for validated benchmark active spaces; CAS(24,18)-scale production trajectories require a future MPS-Krylov response backend.

1 Introduction

Nonadiabatic photochemistry requires electronic-structure methods that remain well behaved near near-degeneracies and conical intersections. CASSCF is a natural reference method in this regime, but the factorial growth of the full-CI active-space solver limits practical applications. DMRG provides a compact MPS representation of large active-space wave functions, but analytic state-averaged DMRG-CASSCF gradients and nonadiabatic couplings are not yet available as a routine open-source PySCF-to-dynamics workflow.

This work reports an implementation and validation study. The contribution is not a new response theory; rather, it is a PySCF/pyblock2 implementation, test suite, benchmark characterization, and SHARC interface.

2 Theory

We start from the state-averaged CASSCF Lagrangian and coupled-perturbed orbital-response equations. In the DMRG variant, active-space CI vectors are represented by MPS wave functions and response quantities are evaluated either through a native MPS response path or through a PySCF-compatible wrapper that exposes the expected FCI-solver contract for validation.

2.1 Bond-Dimension Orbital-Response Error

For a fixed set of FCI-optimized CASSCF orbitals, we replace the active-space FCI vector by a bond-dimension- M DMRG approximation and recompute analytic gradients and nonadiabatic couplings. This isolates the bond-dimension orbital-response error from artifacts caused by reoptimizing orbitals on an incompletely converged truncated CI manifold.

3 Implementation

The code provides two validated paths:

- `CPDMRG_CASSCF_ResponseMPS`: native MPS response operations.
- `MPSAsFCISolver`: a PySCF-compatible active-space solver wrapper used by the current production validation path.

The SHARC interface adds a non-hybrid electronic-structure method name, `dmrg-casscf`, which installs the DMRG-backed active-space solver in a PySCF CASSCF object and writes Hamiltonians, dipoles, gradients, and nonadiabatic coupling vectors in SHARC format.

4 Validation

The implementation is validated by unit and integration tests covering FCI/DMRG energies, one- and transition-density matrices, MPS-to-FCI conversion, analytic gradients, analytic nonadiabatic couplings, and SHARC output formatting. The current test report contains 46 passing tests across 14 test files.

Table 1: Public validation systems used in this manuscript draft.

System	Basis	Active space	Role
H ₄ chain	STO-3G	CAS(4,4)	Small-system BVOE validation
H ₂ O	STO-3G	CAS(4,4)	Small-system BVOE validation
N ₂	STO-3G	CAS(6,6)	Medium-CAS BVOE validation
H ₂ O	6-31G	CAS(6,6)	Clean public NAC convergence
C ₂	STO-3G	CAS(8,8)	Large active-space stress test
LiF	STO-3G	CAS(4,4)	Nonzero-NAC stress test with gauge caveat
Ethylene	STO-3G	CAS(2,2)	Public SHARC-interface smoke

5 Results

5.1 BVOE Convergence

Figure 1 shows phase-2 BVOE convergence using real block2 SU2 DMRG at FCI-converged orbitals. The catastrophic spikes observed in the earlier SVD-proxy phase disappear when the response equation is evaluated at stationary FCI-optimized orbitals. H₄, H₂O/STO-3G, N₂, and H₂O/6-31G provide the cleanest convergence story. C₂ supplies a CAS(8,8) stress test, while LiF is retained as a useful but gauge-sensitive nonzero-NAC stress test.

5.2 SHARC Interface Demonstration

The public ethylene smoke test requests Hamiltonians, dipoles, gradients for both singlet states, and the S_0/S_1 derivative coupling. The run completes with SHARC master error code 0 and writes all requested data blocks to `QM.out`. This validates the interface plumbing for method `dmrg-casscf`; it is not presented as a production trajectory claim.

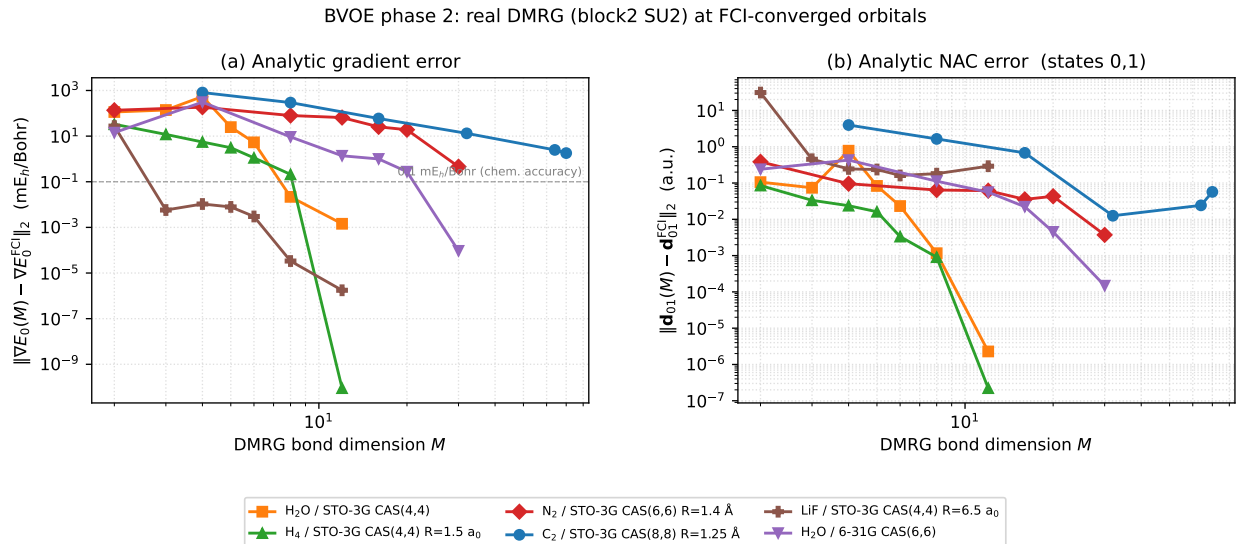


Figure 1: Bond-dimension orbital-response-error convergence for public benchmark systems. Gradient errors are reported in mE_h/Bohr and NAC errors in atomic units. LiF derivative couplings are gauge sensitive and should be interpreted as a stress test rather than the primary NAC convergence proof.

6 Limitations

The current SHARC-facing `dmrg-casscf` path still projects response vectors into FCI-shaped arrays. A determinant-count guard prevents using this path for active spaces where such arrays are not feasible. CAS(24,18)-scale production dynamics will require replacing the FCI-projected response vectors with an MPS-Krylov response backend.

7 Conclusions

We have implemented and validated analytic state-averaged DMRG-CASSCF gradients and nonadiabatic couplings in a PySCF/pyblock2 workflow and added a SHARC-compatible output path. Public benchmark results support manuscript writing now, provided the claims are scoped to implementation, validation, BVOE characterization, and interface demonstration.